



Yakeen Plus (2025)

Practice Test- 10

DURATION :200 Minutes

DATE : 24/11/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (3)
2. (3)
3. (2)
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CHEMISTRY

51. (1)
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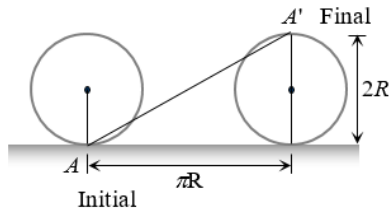
ZOOLOGY

151. (3)
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199. (2)
200. (2)

SECTION-(I) PHYSICS

1. (3)

Horizontal distance covered by the wheel in half revolution = πR .



So the displacement of the point which was initially in contact with ground = $AA' =$

$$\sqrt{(\pi R)^2 + (2R)^2}$$

$$= R\sqrt{\pi^2 + 4} = \sqrt{\pi^2 + 4} \quad (\text{As } R = 1\text{m})$$

2. (3)

Centripetal acceleration

$$= 4\pi^2 n^2 r = 4\pi^2 \left(\frac{1}{2}\right)^2 \times 4 = 4\pi^2$$

3. (2)

$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -4 & 1 \\ 5 & -6 & 6 \end{vmatrix} = -18\hat{i} - 13\hat{j} + 2\hat{k}$$

4. (3)

Case 1: When the man ascends up the rope with acceleration of 2 m/s^2 .

$$\text{We have } T_1 - mg = ma$$

$$\text{or } T_1 = mg + ma$$

$$\text{or } T = 600 + (60 \times 2) = 720 \text{ N}$$

Case 2: When the man descends by a constant velocity.

$$\text{We have } T_2 = mg$$

$$\text{so } T_2 = 50 \times 10 = 500 \text{ N}$$

Case 3: When the man descends by 1 m/s^2 .

$$\text{We have } mg - T_3 = ma$$

$$T_3 = mg - ma$$

$$\text{so } T_3 = 400 - (40 \times 1) = 360 \text{ N}$$

so the net tension at the rigid support is

$$= 720 + 500 + 360 = 1580 \text{ N}$$

5. (2)

Lift is moving up with acceleration a	Apparent weight is greater than true weight
Lift is moving down with acceleration a	Apparent weight is less than true weight
Lift is moving with uniform velocity	Apparent weight is equal to true weight
Lift is freely falling	Apparent weight is zero

6. (2)

$$a = \frac{m_2}{m_1 + m_2} g$$

$$= \frac{3}{7+3} 10 = 3 \text{ m/s}^2$$

7. (3)

$$T_1 = \left(\frac{m_2 + m_3}{m_1 + m_2 + m_3} \right) g = \frac{3+5}{2+3+5} \times 10 = 8 \text{ N}$$

8. (3)

The tension in the spring will be the reading of dynamometer.

$$F_{\text{ext}} = Ma$$

$$50 - 30 = 10(a)$$

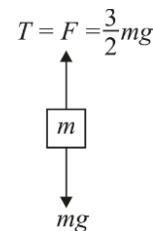
$$a = 2 \text{ m/s}^2$$

for 6 kg block

$$50 - T = 6(2)$$

$$T = 38 \text{ N}$$

9. (2)



$$F_{\text{net}} = ma$$

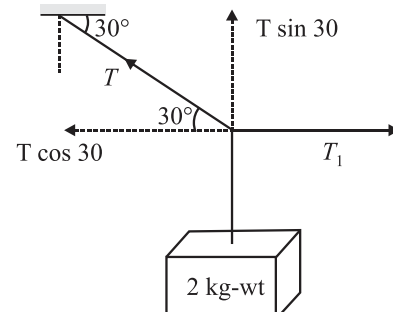
$$\frac{3}{2}mg - mg = ma$$

$$a = \frac{g}{2}$$

10. (3)

According to Newton's first law motion.

11. (3)



$$T \sin 30 = 2 \text{ kg wt}$$

$$\Rightarrow T = 4 \text{ kg wt}$$

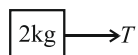
$$T_1 = T \cos 30^\circ$$

$$= 4 \cos 30^\circ = 2\sqrt{3}$$

12. (3)

5N force will not produce any tension in spring without support of other 5N force. So here the tension in the spring will be 5N only.

13. (2)



$$F_{\text{net}} = Ma$$

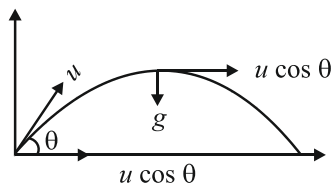
$$(10 \cos 60^\circ) = (3 + 2)a$$

$$a = 1 \text{ m/s}^2$$

$$T = 2(1) = 2 \text{ N}$$

14. (3)

The correct options C $\frac{u^2 \cos^2 \theta}{g}$



We know that, the horizontal component of velocity remains constant and at highest point, the vertical component of velocity is zero. Acceleration will be equal to g and perpendicular to velocity.

At highest point, $mg = \frac{mv^2}{r}$, where r is the radius of curvature

$$\Rightarrow r = \frac{v^2}{g} = \frac{(u \cos \theta)^2}{g}$$

$$\Rightarrow r = \frac{u^2 \cos^2 \theta}{g}$$

15. (3)

$$a = \frac{dv}{dt} = 0.2 \text{ t,}$$

When observe that the acceleration is dependent on time and hence it is non-uniform acceleration.

16. (4)

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$\therefore R \propto u^2$$

It initial velocity be doubled then range will become four times.

17. (4)

$$y = x \tan \theta \left[1 - \frac{x}{R} \right]$$

$$y = 16x - \frac{5x^2}{4}$$

$$= 16x \left[1 - \frac{5x}{4 \times 16} \right]$$

$$= 16x \left[1 - \frac{x}{\left(\frac{4 \times 16}{5} \right)} \right]$$

$$R = \frac{4 \times 16}{5} = \frac{64}{5} = 12.8 \text{ m}$$

18. (3)

$$\omega = \frac{2\pi}{T}$$

$$\frac{\omega_1}{\omega_2} = 1:1$$

19. (3)

Centripetal acceleration $= \frac{v^2}{r}$. It is perpendicular to the rate of increase in speed, i.e., acceleration, which is equal to g according to the question. It is tangential to circular path. Hence, net acceleration

$$\text{of the particle} = \left[\left(\frac{v^2}{r} \right)^2 + g^2 \right]^{1/2}$$

20. (2)

By constrained linear speed of both wheels will be same

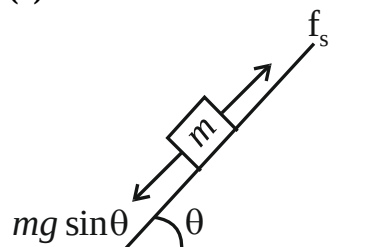
$$\omega_1 r_1 = \omega_2 r_2$$

$$\Rightarrow \omega_2 = \omega_1 \frac{r_1}{r_2}$$

$$= (\alpha t) \frac{r_1}{r_2}$$

$$= \frac{\pi}{4} \times 32 \times \frac{R}{4R} = 2\pi \text{ rad/sec}$$

21. (3)



$$f_s = mg \sin \theta$$

$$f_s = \mu mg \cos \theta = mg \sin \theta$$

$$\tan \theta = \mu = \sqrt{3}$$

$$\theta = 60^\circ$$

22. (3)

At 11th second left is moving upward with acceleration

$$a = \frac{0 - 3.6}{2} = -1.8 \text{ m/s}^2$$

$$\text{Tension in rope } T = m(g+a)$$

$$= 1500 (9.8 - 1.8) = 12000 \text{ N}$$

23. (3)

$F = -\left(\frac{dU}{dx} \right)$ slope of U and x graph, slope is zero at B and C.

24. (4)

$$s = \frac{t^2}{4}; v = \frac{t}{2}$$

$$t = 0, u = 0$$

$$t = 2, v = 1$$

$$\therefore WD = \Delta KE = \frac{1}{2} \times 6 \times 1 = 3J$$

25. (4)

$$F = -\frac{dU}{dx} = -(2x - 6)$$

For equilibrium, $F = 0$

$$x = 3m$$

$$\frac{dU}{dx} = 2x - 6$$

$$\frac{d^2U}{dx^2} = 2 > 0,$$

so $x = 3$ is a point of stable equilibrium.

26. (3)

$$P = \frac{A^3 B^{1/2}}{C^{-4} D^{3/2}} = \frac{A^3 B^{1/2} C^4}{D^{3/2}}$$

$$\frac{\Delta P}{P} \times 100 = 3 \left(\frac{\Delta A}{A} \times 100 \right) + \frac{1}{2} \left(\frac{\Delta B}{B} \times 100 \right) + 4 \left(\frac{\Delta C}{C} \times 100 \right) + \frac{3}{2} \left(\frac{\Delta D}{D} \times 100 \right)$$

27. (1)

$$X = M^a L^b T^c;$$

$$\frac{\Delta X}{X} \times 100 = \left(\frac{a \Delta M}{M} + \frac{b \Delta L}{L} + \frac{c \Delta T}{T} \right) \times 100$$

$$= (a\alpha + b\beta + c\gamma)\%$$

28. (3)

$$F = \frac{\alpha - t^2}{\beta v^2}$$

$$\text{Dimensionally, } \alpha = [T^2]$$

$$[MLT^{-2}] = \frac{[T^2]}{\beta [L^2 T^{-2}]}$$

$$\beta = \frac{T^2}{[MLT^{-2} \cdot L^2 T^{-2}]}$$

$$\Rightarrow \beta = [M^{-1} L^{-3} T^6]$$

Dimensions of

$$\frac{\alpha}{\beta} = \frac{T^2}{M^{-1} L^{-3} T^6} = [ML^3 T^{-4}]$$

29. (2)

$$F = at + bt^2$$

$$F \rightarrow N \rightarrow kg \frac{m}{s^2}$$

$$at \rightarrow N \rightarrow a \cdot t = kg \frac{m}{s^2}; a = kg \frac{m}{s^3};$$

$$[a] = [MLT^{-3}]$$

$$bt^2 \rightarrow N \rightarrow bt^2 = kg \frac{m}{s^2}; b = kg \frac{m}{s^4};$$

$$[b] = [MLT^{-4}]$$

30. (2)

$$\frac{1 \text{ kg m}}{s^2} = \frac{F(x \text{ kg})(y \text{ m})}{(zs)^2}$$

$$\therefore F = \frac{z^2}{xy}$$

31. (3)

$$W = \frac{1}{2}(16+8) \times 4 = 48 \text{ J}$$

32. (3)

$$\text{Let } P_1 = P, \quad P_2 = P_1 + 50\% \quad \text{of} \quad P_1 =$$

$$P_1 + \frac{P_1}{2} = \frac{3P_1}{2}$$

$$E \propto P^2 \Rightarrow \frac{E_2}{E_1} = \left(\frac{P_2}{P_1} \right)^2 = \left(\frac{3P_1/2}{P_1} \right)^2 = \frac{9}{4}$$

$$\Rightarrow E_2 = 2.25E_1 = E_1 + 1.25E_1$$

$$\therefore E_2 = E_1 + 125\% \text{ of } E_1$$

33. (4)

$$\text{Condition for vertical looping } h = \frac{5}{2}r = 5 \text{ cm}$$

$$\therefore r = 2 \text{ cm}$$

34. (2)

$$F = \frac{-\partial u}{\partial x} = -\frac{\partial}{\partial x}(A - Bx^2)$$

$$= -0(0 - 2Bx)$$

$$\Rightarrow F = 2Bx \Rightarrow F \propto x$$

35. (2)

$$W.D = F F_x dx \cos \theta$$

if θ is greater than 90° , then $W.D$ is negative

$\theta < 90^\circ$ $W.D$ is positive.

36. (3)

$$I_{AB} = 1(2)^2 + 1(4)^2$$

$$= 4 + 16 = 20 \text{ kg} \cdot \text{m}^2$$

37. (1)

$$2 \times \frac{1}{3} ml^2 \sin^2 45^\circ$$

38. (3)

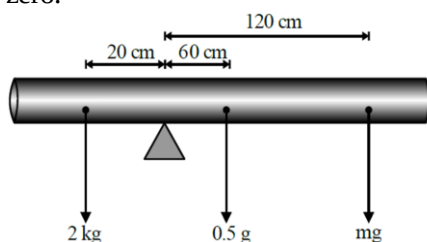
$$I = \frac{MR^2}{2}$$

$$\text{but } 2\pi R = L \Rightarrow R = \frac{L}{2\pi}$$

$$\therefore I = \frac{ML^2}{8\pi^2}$$

39. (4)

For the rod to be in equilibrium, net torque due to gravity forces acting on the rod about pivot must be zero.



Balancing torque about pivot, we have

$$2 \text{ g} \times 20 = 0.5 \text{ g} \times 60 + mg \times 120$$

$$\Rightarrow m = \frac{0.5}{6} \text{ kg} = \frac{1}{12} \text{ kg}$$

40. (1)

Given $I_{\text{Hollow cylinder}} = I_{\text{Solid cylinder}}$

$$\Rightarrow MR_1^2 = \frac{1}{2}MR_2^2$$

$$\therefore \frac{R_1}{R_2} = \frac{1}{\sqrt{2}}$$

41. (3)

$$O = m_1\vec{u}_1 + m_2\vec{u}_2$$

$$m_1\vec{u}_1 = m_2\vec{u}_2$$

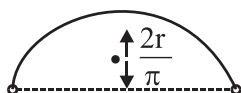
$$m_1 = m_2$$

$$\vec{u}_1 = \vec{u}_2$$

42. (4)

$$I = 2 \left[\frac{MR^2}{4} + MR^2 \right]$$

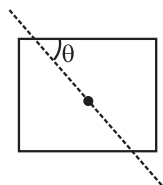
43. (3)



$$mr^2 = I_{cm} + m \left(\frac{2r}{\pi} \right)^2$$

$$I_{cm} = mr^2 \left[1 - \frac{4}{\pi^2} \right]$$

44. (3)



Using perpendicular axis theorem,

$$I_z = I_x + I_y$$

$$I = 2I'$$

$$I' = \frac{I}{2} = \text{Constant}$$

45. (1)

$$P_i = P_f$$

$$Mx = m(L - x)$$

$$x = \frac{mL}{M + m}$$

46. (2)

$$\omega = 10t^2 + 15$$

$$\alpha = \frac{d\omega}{dt} = 20t + 0$$

$$\alpha_{(t=1 \text{ s})} = 20 \text{ rad/s}^2$$

47. (3)

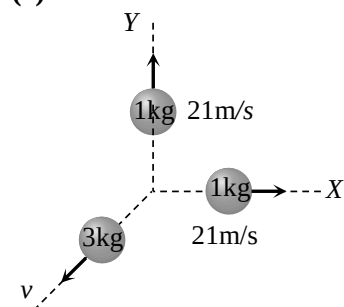
$$\alpha = \frac{\tau}{I}$$

$$\alpha = \frac{10}{100} = 0.1 \text{ rad/s}^2$$

48. (3)

$$m_G = \frac{m_B v_B}{v_G} = \frac{50 \times 10^{-3} \times 30}{1} = 1.5 \text{ kg}$$

49. (4)



$$P_x = m \times v_x = 1 \times 21 = 21 \text{ kg m/s}$$

$$P_y = m \times v_y = 1 \times 21 = 21 \text{ kg m/s}$$

$$\therefore \text{Resultant} = \sqrt{P_x^2 + P_y^2} = 21\sqrt{2} \text{ kg m/s}$$

The momentum of heavier fragment should be numerically equal to resultant of $\vec{P}_x$ and $\vec{P}_y$.

$$3 \times v = \sqrt{P_x^2 + P_y^2} = 21\sqrt{2} \therefore v = 7\sqrt{2} = 9.89 \text{ m/s}$$

50. (4)

$$I' = I_{\text{COM}} + Md^2$$

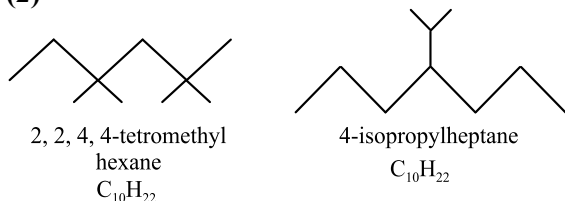
$$\Rightarrow MK^2 = \frac{Ml^2}{12} + Md^2$$

$$\Rightarrow \frac{Ml^2}{4} - \frac{Ml^2}{12} = Md^2 \Rightarrow d = \frac{1}{\sqrt{6}}$$

SECTION-(II) CHEMISTRY

51. (1)
C¹².
[New NCERT Class 11th Page No. 9]
52. (1)
Molecular formula of methane = CH₄
Empirical formula of methane = CH₄
Molecular formula of ethane = C₂H₆
Empirical formula of ethane = CH₃
Molecular formula tells us how many atoms of each element are in the compound, and empirical formulas tells us the simplest or most reduced ratio of elements in a compound. If a compound's molecular formula cannot be reduced any more, then the empirical formula is same as molecular formula.
[New NCERT Class 11th Page No. 20]
53. (4)
C₃H₆O + 4O₂ → 3CO₂ + 3H₂O
1 volume give 3 volume of CO₂
40 mL gives 3 × 40 mL of CO₂ = 120 mL
[New NCERT Class 11th Page No. 18]
54. (1)
H₂SO₄ + Ca(OH)₂ → CaSO₄ + 2H₂O
1 mole of H₂SO₄ reacts with 1 mole of Ca(OH)₂ to give 1 mole of CaSO₄.
Thus, Ca(OH)₂ is limiting reagent. Hence, 0.2 mole of CaSO₄ is formed.
[New NCERT Class 11th Page No. 17]
55. (1)
A < C < B < D.
[New NCERT Class 11th Page No. 10]
56. (4)
Pb(82) is a member of p-block.
[New NCERT Class 11th Page No. 84]
57. (4)
32 elements.
[New NCERT Class 11th Page No. 118]
58. (1)
1 mole of O₂(g) at STP contains Avogadro number (N_A = 6.022 × 10²³) of O₂ molecules.
[New NCERT Class 11th Page No. 120]
59. (2)
Molality = 3 m = 3 mol/kg
 $n_{\text{NaOH}} = 3, n_{\text{H}_2\text{O}} = \frac{1000}{18} = 55.55$
 $\chi_{\text{NaOH}} = \frac{n_{\text{NaOH}}}{n_{\text{NaOH}} + n_{\text{H}_2\text{O}}} = \frac{3}{58.55}$
 $\chi_{\text{NaOH}} = 0.05$
So, $\chi_{\text{H}_2\text{O}} = 0.95$
[New NCERT Class 11th Page No. 121]
60. (1)
Vapour pressure is pressure exerted by vapour in equilibrium with liquid.
[New NCERT Class 11th Page No. 119]
61. (2)
Gas deviates from ideal behaviour at high pressure and low temperature.
[New NCERT Class 11th Page No. 20]
62. (3)
According to Charles' law, the volume occupied by fix amount of gas is directly proportional to the absolute temperature of the gas at a constant pressure.
[New NCERT Class 11th Page No. 11]
63. (4)
Uue ⇒ 119
[New NCERT Class 11th Page No. 13]
64. (3)
On moving down the group from F to I then, The increase in electron density will result in larger ions.
As atomic size and nuclear charge both rise, ionisation energy will decrease.
The ability to easily remove electrons will result in an increase in oxidising capability. As nuclear charge drops, electronegativity decreases (nuclear force of attraction)
Hence, correct options are ii, iii, iv.
[New NCERT Class 11th Page No. 9]
65. (2)
F < O < C < Cl < Br.
[New NCERT Class 11th Page No. 15]
66. (3)
Acetone when mixed with chloroform form H-bonding which lead to decreases in vapour pressure, hence, show negative deviation.
[New NCERT Class 11th Page No. 120]
67. (2)
The depression in freezing point is given as:
 $\Delta T_f = i \times K_f \times m$
m = molality
 $m = \frac{5 \times 1000}{142 \times 45} = 0.782$
 $\Delta T_f = i \times K_f \times m$
 $i = \frac{\Delta T_f}{K_f \times m} = \frac{3.82}{1.86 \times 0.782} = 2.63$
[New NCERT Class 11th Page No. 123]
68. (4)
Isomers should have same molecular formula.
[New NCERT Class 11th Page No. 119]

69. (2)



[New NCERT Class 11th Page No. 18]

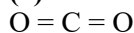
70. (4)

Ambident nucleophiles are the species which contain at least two donor sites.

Examples: NH_2 — OH , $NCO^\ominus$, $NO_2^\ominus$

[New NCERT Class 11th Page No. 19]

71. (2)



[New NCERT Class 11th Page No. 24]

72. (2)

$$\Delta T_b = K_b \times m$$

$$0.52 = 0.52 \times m$$

$$m = 1$$

$$m = \frac{w \times 1000}{M_x \times W} = \frac{6}{M_x} \times \frac{1000}{100} = 1$$

Molar mass of X (M_x) = 60 g mol⁻¹

[New NCERT Class 12th Page No. 17]

73. (1)

$$\pi = iCRT$$

$$\pi = \frac{i n}{V} RT ; \pi V = i n RT$$

[New NCERT Class 12th Page No. 24]

74. (2)

According to Henry's law, the solubility of a gas in a given volume of liquid increases with an increase in pressure.

[New NCERT Class 11th Page No. 25]

75. (3)

n-propyl amine, isopropyl amine, trimethyl amine, ethyl methyl amine

[New NCERT Class 11th Page No. 22]

76. (2)

$$\therefore \Delta T_b = K_b \times m \Rightarrow K_b = \frac{\Delta T_b}{m}$$

[New NCERT Class 12th Page No. 18]

77. (3)

$$\frac{\Delta P}{P^o} \propto \chi_{\text{solute}}$$

[New NCERT Class 11th Page No. 19]

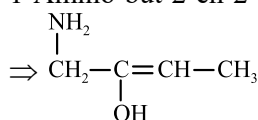
78. (4)

Both benzene and toluene are non-polar aromatic hydrocarbons. They have similar sizes and shapes, and their non-polar nature promotes mixing without specific interactions disrupting the ideal behavior. This makes an ideal solution. The interaction between molecules of benzene and toluene will be the same as the interaction between their own molecules.

[New NCERT Class 12th Page No. 13]

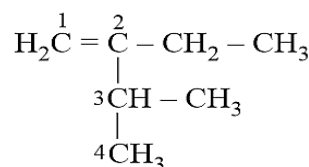
79. (3)

1-Amino-but-2-en-2-ol



[New NCERT Class 11th Page No. 14]

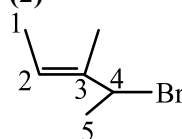
80. (1)



2-ethyl-3-methyl but-1-ene

[New NCERT Class 11th Page No. 16]

81. (2)



[New NCERT Class 11th Page No. 17]

82. (2)



[New NCERT Class 11th Page No. 20]

83. (2)

Due to absence of d-orbital in oxygen. OF_4 does not exist.

[New NCERT Class 11th Page No. 21]

84. (1)



[New NCERT Class 11th Page No. 9]

85. (3)

$$\text{molality} = \frac{\text{no. of mole} \times 1000}{\text{mass of solvent (g)}}$$

$$\text{Mass of solvent} = \frac{0.5 \times 1000}{1} = 500g$$

[New NCERT Class 11th Page No. 11]

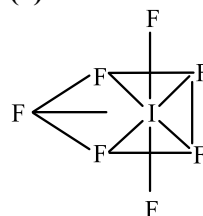
86. (4)

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 (\pi 2p_x^2 = \pi 2p_y^2) \sigma 2p_z^1$$

Now from the electronic configuration we can see that the number of electrons in the $\sigma 2p$ molecular orbital is 1.

[New NCERT Class 11th Page No. 10]

87. (3)



[New NCERT Class 11th Page No. 11]

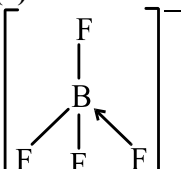
88. (3)
Both NF_3 and NH_3 have identical shapes and a lone pair of electrons on nitrogen. The dipole moment of NH_3 is higher than NF_3 due to different direction of moments of the N-F bonds. BF_3 is a symmetrical molecule. It has zero dipole moment.
[New NCERT Class 11th Page No. 14]

89. (2)
1 molecule of $\text{Ca}(\text{OH})_2$ contains 5 atoms,
 $\therefore$ 1 mole of $\text{Ca}(\text{OH})_2$ contains 5 mole atoms.
 $\therefore 5 \times 6.022 \times 10^{23}$ atom/mol.
[New NCERT Class 11th Page No. 20]

90. (3)
Given that an element with electronic configuration $[\text{Ar}]3d^1$ in its +2 oxidation state.
The atomic number of element = $\text{Ar} + 1 + 2 = 21$
Thus, the element is Scandium. It belongs to 4th period and group-3
[New NCERT Class 11th Page No. 21]

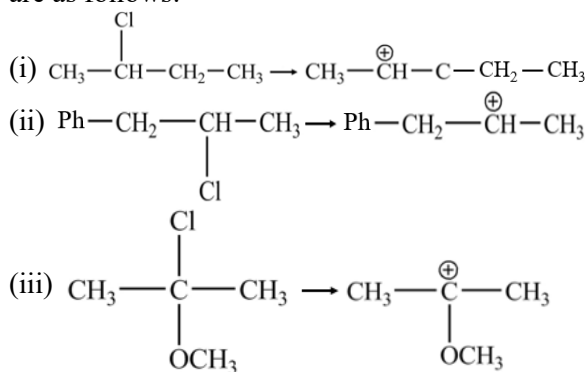
91. (4)
No. of moles = $\frac{16}{32} = \frac{1}{2}$
No. of atoms = $\frac{1}{2} \times N_A \times 2 = N_A = 6.022 \times 10^{23}$
[New NCERT Class 11th Page No. 24]

92. (1)
Zeros at the beginning of a number are not significant.
[New NCERT Class 11th Page No. 25]

93. (2)

[New NCERT Class 11th Page No. 114]

94. (4)
 CH_3^+ , AlCl_3 , H_3O^+ are electrophiles.
[New NCERT Class 11th Page No. 120]

95. (3)
The rate of $\text{S}_{\text{N}}1$ reaction depends on the stability of carbocation intermediate. The stability of carbocation is directly proportional to +I, +M and H-effect. The carbocation intermediates formed are as follows:



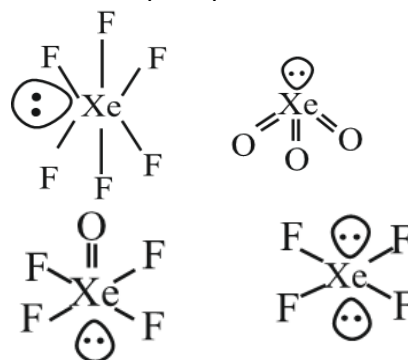
(iv) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl} \rightarrow \text{CH}_3-\text{CH}_2-\text{CH}_2^+$
The general stability order of carbocation is Primary < Secondary < Tertiary. Except (iv) which is a primary carbocation all the other carbocations are secondary hence (iv) will be least stable. The carbocation formed in case of (i) and (ii) are stabilized by hyperconjugation but additionally the carbocation in (ii) is stabilized by resonance through benzene ring. In case of (iii) the carbocation is stabilized by +M-effect of $-\text{OCH}_3$ group which is more efficient as compared to the benzene ring.

Therefore, the stability order of carbocation is $\text{iii} > \text{ii} > \text{i} > \text{iv}$.

Therefore, the correct reactivity order is as follows $\text{iii} > \text{ii} > \text{i} > \text{iv}$

[New NCERT Class 11th Page No. 122]

96. (1)
 $\text{XeF}_6 \Rightarrow$ Distorted octahedral
 $\text{XeO}_3 \Rightarrow$ Pyramidal
 $\text{XeOF}_4 \Rightarrow$ Square pyramidal
 $\text{XeF}_4 \Rightarrow$ Square planar



[New NCERT Class 11th Page No. 123]

97. (2)
 $\text{Mg}^+ = 1s^2 2s^2 2p^6 3s^1$
 $\text{Na}^+ = 1s^2 2s^2 2p^6$
Second I. E. of Na is more due to fully filled 2p orbital.
[New NCERT Class 11th Page No. 120]

98. (1)
Increases in average molecular speed.
[New NCERT Class 11th Page No. 23]

99. (1)
Due to dissociation in NaCl , Raoult's law is not applicable.
[New NCERT Class 11th Page No. 119]

100. (2)
 $\text{Ethanol} + \text{H}_2\text{O} \rightarrow$ Azeotropic mixture, Non-ideal solution with positive deviation.
Ebullioscopic constant (K_b) $\Rightarrow$ Molal elevation constant.
Cryoscopic constant (K_f) $\Rightarrow$ Molal depression constant
 $\text{H}_2\text{O} + \text{H}_2\text{SO}_4 \rightarrow$ Azeotropic mixture, Non-ideal solution with negative deviation.
[New NCERT Class 12th Page No. 18]

SECTION-(III) BOTANY

101. (3)

Saprophyte	Decomposition of dead organic materials
Parasite	Depend on living plants or animals
Lichens	Symbiotic association of algae and fungi
Mycorrhiza	Symbiotic association of fungi with plant roots

[New NCERT Class 11th Page No. 16]

102. (1)

Generic name starts with capital letter and specific epithet should be written starting with small letters.

[New NCERT Class 11th Page No. 4]

103. (1)

Fusion between one large, nonmotile (static) female gamete and a smaller, motile male gamete is termed oogamous, e.g., *Volvox*, *Fucus*.

[New NCERT Class 11th Page No. 24]

104. (1)

Petunia – Dicotyledonae

[New NCERT Class 11th Page No. 7,8]

105. (2)

The asexual spores are conidia produced exogenously on the special mycelium called conidiophores.

Sexual spores are called ascospores which are produced endogenously in sac like asci (singular ascus). These asci are arranged in different types of fruiting bodies called ascocarps.

[New NCERT Class 11th Page No. 18]

106. (3)

Carnivora is order and it is a group of organisms belonging to related families

[New NCERT Class 11th Page No. 7]

107. (4)

Yeast is unicellular

[New NCERT Class 11th Page No. 17,18]

108. (1)

The thalloid plant body of liverwort is dorsiventrally appressed closely to substrate.

[New NCERT Class 11th Page No. 29]

109. (4)

- In prokaryotes, ribosomes are associated with the plasma membrane of the cell.
- Several ribosomes on a single mRNA form a structure called polysome.
- Polysome translates the mRNA into proteins.
- Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.

[New NCERT Class 11th Page No. 91]

110. (3)

In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant.

[New NCERT Class 11th Page No. 88]

111. (3)

As we go higher from species to kingdom, the number of common characteristics goes on decreasing. Lower the taxa, more are the characteristics that the members within the taxon share. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

[New NCERT Class 11th Page No. 8]

112. (3)

Nostoc and *Anabaena*-Monera

Amoeba, *Entamoeba* -Protista

Ustilago, *Claviceps*, *Colletotrichum*-Fungi

[New NCERT Class 11th Page No. 13,16,18]

113. (1)

‘Animals’, ‘mammals’ and ‘dogs’ represent taxa at different levels.

[New NCERT Class 11th Page No. 5]

114. (4)

The endomembrane system includes endoplasmic reticulum (ER), Golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

[New NCERT Class 11th Page No. 95]

115. (4)

During anaphase I the homologous chromosomes separate, while sister chromatids remain associated at their centromeres.

[New NCERT Class 11th Page No. 127]

116. (4)

Metacentric chromosome	Middle centromere forming two equal arms of the chromosome
Sub-metacentric chromosome	Centromere slightly away from the middle of the chromosome
Acrocentric chromosome	Centromere is situated close to chromosome end forming one extremely short and one very long arm
Telocentric chromosome	Terminal centromere on the chromosome

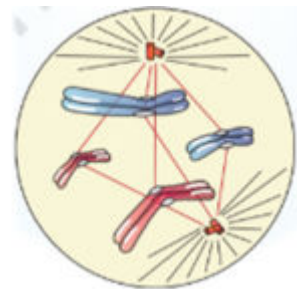
[New NCERT Class 11th Page No. 101]

117. (4)
Mesosome is present in bacteria.
[New NCERT Class 11th Page No. 90]
118. (3)
Phosphate granules are example of inclusion bodies.
Cyanophycean granules and glycogen granules are not bound by any membrane.
[New NCERT Class 11th Page No. 91]
119. (3)
Cell is the fundamental structural and functional unit of all non-living organisms.
[New NCERT Class 11th Page No. 87]
120. (2)
Brinjal belongs to the genus *Solanum*.
[New NCERT Class 11th Page No. 7]
121. (4)
The average duration of cell cycle of a human cell is 24 hour.
[New NCERT Class 11th Page No. 121]
122. (2)
 - Mycoplasma can survive without oxygen.
 - Mycoplasma completely lack a cell wall.
[New NCERT Class 11th Page No. 14]
123. (4)
Class-Mammalia
[New NCERT Class 11th Page No. 8]
124. (1)
Gemmae are green, multicellular, asexual buds.
[New NCERT Class 11th Page No. 29]
125. (3)
Oogamous and non-motile gamete with chlorophyll *a* present in algae *Porphyra*.
[New NCERT Class 11th Page No. 27, 28]
126. (4)
During metaphase chromosomes are arranged on the equator of spindle.
[New NCERT Class 11th Page No. 91]
127. (3)
The walls of diatoms are embedded with silica and thus the walls are indestructible.
[New NCERT Class 11th Page No. 14]
128. (4)
Mature heart cells cannot undergo cell division. These cells are probably considered in G₀ phase.
[New NCERT Class 11th Page No. 122]

129. (4)
 - Plants show continuous growth throughout their life.
 - Mitosis occurs in apical meristematic cells.
 - A very significant contribution of mitosis is cell repair.
 - Lateral cambium divides by mitosis.
[New NCERT Class 11th Page No. 125]

130. (4)
International Code of Zoological Nomenclature.
[New NCERT Class 11th Page No. 4]

131. (3)



It represents late prophase. After this metaphase occurs.

[New NCERT Class 11th Page No. 123]

132. (1)
A→D→B→C
A. *Petunia*
D. Solanaceae
B. Polymoniales
C. Dicotyledonae
[New NCERT Class 11th Page No. 7, 8]

133. (3)
The second stage is the leafy stage, which develops from the secondary protonema as a lateral bud. They consist of upright, slender axes bearing spirally arranged leaves. They are attached to the soil through multicellular and branched rhizoids. This stage bears the sex organs.
[New NCERT Class 11th Page No. 30]

134. (2)
The sex organs in bryophytes are multicellular. Male sex organ is called antheridium. They produce biflagellate antherozoids.
[New NCERT Class 11th Page No. 29]

135. (4)
Order-Primata and Diptera
[New NCERT Class 11th Page No. 8]

136. (2)
Anaphase occurs in mitosis.
[New NCERT Class 11th Page No. 122]

137. (1)

Slime layer	Loose sheath
Capsule	Thick and tough
Cell wall	Strong structural support
Cell membrane	Interacts with the outside world

[New NCERT Class 11th Page No. 90]

138. (2)

The sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it.

[New NCERT Class 11th Page No. 29]

139. (2)

Unlike bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes do not have an independent free-living existence.

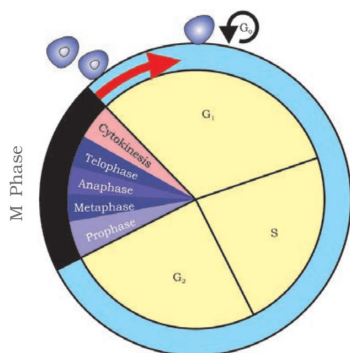
[New NCERT Class 11th Page No. 33]

140. (4)

Taxonomy takes into account characterisation, identification, nomenclature and classification.

[New NCERT Class 11th Page No. 5]

141. (1)



A-G₁ phase

B-S phase

C-G₂ phase

D-M phase

[New NCERT Class 11th Page No. 121]

142. (3)

The outer membrane of nucleus usually remains continuous with the endoplasmic reticulum and also bears ribosomes on it.

[New NCERT Class 11th Page No. 100]

143. (4)

Lion belongs to Felidae family.

[New NCERT Class 11th Page No. 7]

144. (4)

The small bristle like fibres sprouting out of the bacterial cell are fimbriae.

[New NCERT Class 11th Page No. 91]

145. (2)

Canidae	Family
Sapindales	Order
Insecta	Class
Arthropoda	Phylum

[New NCERT Class 11th Page No. 8]

146. (2)

- The chromosome becomes gradually visible with compaction of chromatin-Leptotene
- The complex formed by a pair of synapsed homologous chromosomes-Zygotene
- Recombination nodules are seen- Pachytene

[New NCERT Class 11th Page No. 126]

147. (4)

The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.

It also contains small, double stranded circular DNA molecules and ribosomes.

[New NCERT Class 11th Page No. 93]

148. (1)

Chemotaxonomy uses the chemical constituents of the plant.

[New NCERT Class 11th Page No. 24]

149. (1)

- The lipids are arranged within the cell membrane with the Polar heads towards the outer sides and the hydrophobic tails towards the inner part.
- Fluid mosaic model of the structure of cell membrane was proposed by Singer and Nicolson (1972).

[New NCERT Class 11th Page No. 93, 94]

150. (3)

Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.

It is nearly impossible to study all the living organisms.

[New NCERT Class 11th Page No. 5]

SECTION-(IV) ZOOLOGY**151. (3)**

Thymine pairs with adenine in a DNA molecule.

[New NCERT Class 11th Page No. 109]**152. (1)**

Digestive enzyme milk, saliva, earwax, milk and oil are secreted by exocrine glands.

[Old NCERT Class 11th Page No. 102]**153. (2)**

- Dense regular connective tissue consists of ligaments and tendons.
- Tendons connect skeletal muscles to bones and ligaments connects bone to bone.

[Old NCERT Class 11th Page No. 103]**154. (2)**

Starch is a carbohydrate.

[New NCERT Class 11th Page No. 110]**155. (2)**

Malpighian tubules	100 – 150
Hepatic caeca	6 – 8
Forewings	Opaque dark
Hindwings	Used in flight

[Old NCERT Class 11th Page No. 114]**156. (1)**

Statocysts or balancing organs are present in arthropods.

[New NCERT Class 11th Page no. 44]**157. (2)**

- The entire body of cockroach, is covered by a hard chitinous exoskeleton.
- In each segment, exoskeleton has hardened plates called sclerites that are joined to each other by a thin and flexible articular membrane or arthrodial membrane.

[Old NCERT Class 11th Page No. 112]**158. (2)**

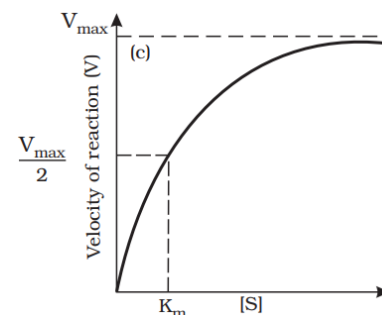
Blood	White blood cells
Lacunae	Osteocytes
Cartilage	Outer ear joints
Fibroblast	Areolar tissue

[Old NCERT Class 11th Page No. 103]**159. (2)**

- Adipose tissues are specialised to store fats and extra nutrients which are not used immediately by the body get converted into fats.
- Adipose tissue is a type of loose connective tissue.

[Old NCERT Class 11th Page No. 103]**160. (2)**

- Tight junctions help to stop substances from leaking across a tissue.
- Adhering junctions perform cementing to keep neighbouring cells together.
- Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

[Old NCERT Class 11th Page No. 102]**161. (3)**

Michaelis constant (K_m) value of an enzyme is the substrate concentration at which the velocity of reaction reaches $\frac{1}{2} V_{\max}$.

[New NCERT Class 11th Page No. 116]**162. (1)**

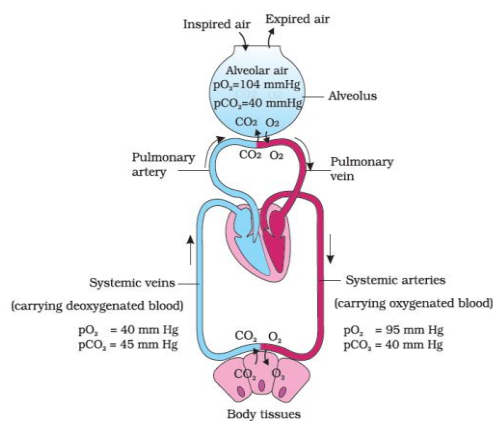
In frogs, fertilisation is external and takes place in water.

[Old NCERT Class 11th Page No 120]**163. (4)**

- Neurons are the structural and functional unit of neural system and are excitable cells.
- They exert greatest control over body's responsiveness to changing condition.

[Old NCERT Class 11th Page No. 105]

164. (3)



Systemic veins carry deoxygenated blood to heart and having pO_2 is 40mmHg.

[New NCERT Class 11th Page no. 187]

165. (2)

Phosphodiester bond is present between nucleotides in a nucleic acid.

[New NCERT Class 11th Page No. 112]

166. (1)

Long exposure to dust can cause occupational respiratory disorders.

[New NCERT Class 11th Page no. 188]

167. (1)

Fishes	Through gills
Insects	Through a network of tracheal tubes
Human	Through lungs
Amphibians	Through skin

[New NCERT Class 11th Page no. 184]

168. (4)

- The outer pleural membrane is in close contact with the thoracic lining.
- The inner pleural membrane is in contact with the lung surface.
- The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part.
- The maximum volume of air a person can breathe in after a forced expiration is the vital capacity.

[New NCERT Class 11th Page no. 185]

169. (2)

Sinus venosus collects deoxygenated blood and joins the right atrium.

[New NCERT Class 11th Page no. 118]

170. (4)

- Glucose is not found in acid-insoluble pool.
- The acid insoluble fraction, has four types of organic compounds i.e., proteins, nucleic acids, polysaccharides and lipids.

[New NCERT Class 11th Page No. 105]

171. (1)

Trachea is a straight tube extending up to the mid-thoracic cavity which divides at the level of 5th thoracic vertebrae.

[New NCERT Class 11th Page no. 184]

172. (2)

Every 100 mL of oxygenated blood delivers approximately 5 mL of O_2 to the tissues.

[New NCERT Class 11th Page no. 189]

173. (3)

Neural signal from pneumotaxic centre can reduce the duration of inspiration.

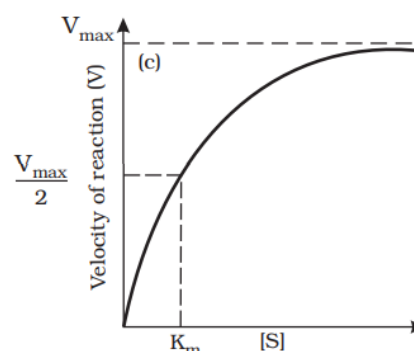
[New NCERT Class 11th Page no. 189]

174. (2)

Contraction of external intercostal muscles increases the volume of thoracic chamber in the dorsoventral axis.

[New NCERT Class 11th Page no. 186]

175. (1)



Concentration of substrate on enzyme activity

[New NCERT Class 11th Page No. 116]

176. (4)

Chordata, Arthropoda and Annelida are triploblastic, metamerically segmented and coelomate animals.

[New NCERT Class 11th Page no. 43]

177. (3)

- The body of Hemichordates is divided into three regions: proboscis, collar, and trunk.
- Hemichordata consists of worm like marine animals.

[New NCERT Class 11th Page no. 44]

178. (3)

- Echinodermates are marine with organ-system level of organisation.
- Hemichordates comprises of small group of wormlike marine animals with organ-system level of organisation.

[New NCERT Class 11th Page no. 45]

179. (2)

- *Aplysia*, *Loligo*, *Sepia* are Mollusca.
- *Cucumaria* is the example of phylum Echinodermata.

[New NCERT Class 11th Page no. 44]

180. (4)

Aschelminthes are triploblastic, pseudocoelomate with organ system level of organisation.

[New NCERT Class 11th Page no. 46]

181. (4)

Nereis are coelomates.

[New NCERT Class 11th Page no. 43]

182. (2)

- *Pteropus* is a viviparous mammal.
- *Echinus* is echinodermata.

[New NCERT Class 11th Page no. 42]

183. (4)

Arthropods have malpighian tubules for excretion.

[New NCERT Class 11th Page no. 44]

184. (3)

All of the following are characteristics of the phylum arthropoda:

- Balancing organs are present.
- Body is covered by a chitinous exoskeleton.
- Body consists of head, thorax and abdomen.
- Circulatory system is of open type.

[New NCERT Class 11th Page no. 44]

185. (2)

Aurelia belongs to phylum coelenterata.

[New NCERT Class 11th Page no. 41]

186. (4)

- Closed type of blood circulation is shown by: *Nereis*, *Pheretima*, *Naja*.
- Open type of blood circulation is shown by: *Laccifer*, *Limulus*, *Aedes* mosquito, *Saccoglossus*

[New NCERT Class 11th Page no. 44]

187. (4)

- Female cockroach possesses sixteen ovarioles in the ovaries.
- Cockroaches exhibit mosaic vision with more sensitivity and less resolution.
- A mushroom-shaped gland is present in the 6th–7th abdominal segments of male cockroach.

[Old NCERT Class 11th Page No. 114]

188. (1)

External fertilisation occurs in class Amphibia.

[New NCERT Class 11th Page no. 48]

189. (3)

In a polysaccharide chain (say glycogen), the right end is called the reducing end and the left end is called the non-reducing end.

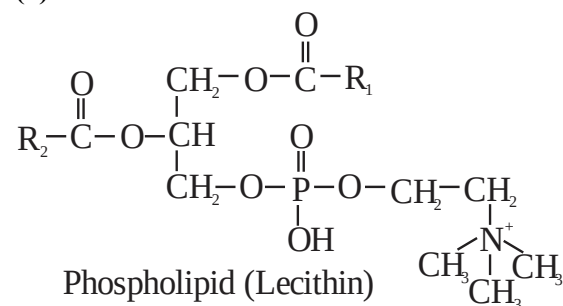
[New NCERT Class 11th Page No. 110]

190. (1)

- Frog have the ability to change colour.
- This protective coloration is called mimicry.

[Old NCERT Class 11th Page No. 116]

191. (4)



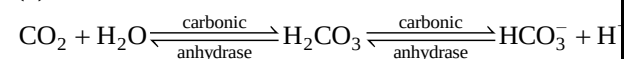
[New NCERT Class 11th Page No. 107 Fig: 9.1]

192. (1)

- Glycogen is a polymer of glucose.
- The complex polysaccharides are mostly homopolymers.

[New NCERT Class 11th Page No. 113]

193. (1)



- Carbon dioxide entering erythrocytes reacts with water to form carbonic acid. The enzyme involved is carbonic anhydrase.

[New NCERT Class 11th Page no. 190]

194. (1)

Vinblastine is a drug.

[New NCERT Class 11th Page No. 107]

195. (2)

The tracheae, primary, secondary and tertiary bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.

[New NCERT Class 11th Page no. 184]

196. (3)

Apoenzyme	Protein part of the enzyme.
NADP	Niacin
Inhibitor of succinate dehydrogenase	Malonate
Haemoglobin	Quaternary structure of protein

[New NCERT Class 11th Page No. 117]

197. (4)

- Squamous epithelium is found in the walls of blood vessels and alveoli.
- Columnar epithelium are found in the stomach and intestine.
- Compound epithelium are found in moist surface of buccal cavity and pharynx.

[Old NCERT Class 11th Page No. 101]

198. (2)

- Connective tissues are most abundant in the body.
- Blood is an example of specialised connective tissue.

[Old NCERT Class 11th Page No. 102]

199. (2)

- Rate of the chemical reaction doubles on increasing the temperature by 10°.
- Catalysed reactions proceed at rates higher than that of uncatalysed reactions.

[New NCERT Class 11th Page No. 110]

200. (2)

Expiratory capacity = Tidal volume + Expiratory reserve volume

$$EC = TV + ERV$$

$$EC = 700 + 1000 = 1700 \text{ mL}$$

[New NCERT Class 11th Page no. 185]



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